

Working with Data

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1 library & Load

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4    v readr      2.1.5
v forcats    1.0.0    v stringr    1.5.1
v ggplot2    3.5.1    v tibble     3.2.1
v lubridate  1.9.4    v tidyr      1.3.1
v purrr      1.0.4
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(descr)
library(knitr)
library(dplyr)
library(Hmisc)
```

Attaching package: 'Hmisc'

The following objects are masked from 'package:dplyr':

src, summarize

The following objects are masked from 'package:base':

format.pval, units

```
library(readr)
library(readxl)
library(ggplot2)
library(car)
```

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

The following object is masked from 'package:purrr':

some

```
Project_Data = read.csv("/cloud/project/Data/Connection_to_Nature_Data.csv",
header = TRUE)
```

2 Single Variables

```
# VARIABLE 1: People's Age

# Age 18 and above

Project_Data <- subset(Project_Data, D_Age >= 18)

# Group D_Age into 4 age range's and Labeling

Project_Data$Age_Group <- cut(Project_Data$D_Age,
                              breaks = c(18, 25, 40, 65, Inf),
                              labels = c("18-25", "26-40", "40-65", "65+"),
                              right = TRUE,
                              include.lowest = TRUE)

# Frequency table

freq(as.ordered(Project_Data$Age_Group), plot = FALSE)
```

```
as.ordered(Project_Data$Age_Group)
  Frequency Percent Cum Percent
18-25      49   9.515      9.515
26-40      85  16.505     26.019
40-65     301  58.447     84.466
65+       80  15.534    100.000
Total     515 100.000
```

I chose this variable (**age**) because I think it would be important to look at in reference to how loneliness and time spent in nature varies among age groups. Perhaps depending on the age group, there will be more positive benefits to those exposed to nature in relation to loneliness.

```
# VARIABLE 2: How many hours on average do you currently spend in nature per week?

# Group D_hours into 4 categories & Labeling

Project_Data$Nature_Hours_Group <- cut(Project_Data$D_hours,
                                       breaks = c(0, 5, 15, 30, Inf),
                                       labels = c("Low (0-5)", "Moderate (6-15)", "High (16-30)", "Very High (30+)"),
                                       include.lowest = TRUE)

# Frequency table

freq(as.ordered(Project_Data$Nature_Hours_Group), plot = FALSE)
```

```
as.ordered(Project_Data$Nature_Hours_Group)
```

	Frequency	Percent	Cum Percent
Low (0-5)	137	26.602	26.60
Moderate (6-15)	230	44.660	71.26
High (16-30)	118	22.913	94.17
Very High (30+)	30	5.825	100.00
Total	515	100.000	

This is very important. This variable (**hours spent in nature**) is important because when I did the literature review assignment, depending on the time spent in nature, actually lowered both social loneliness and emotional loneliness, but it depends how much time was spent in nature.

```
# VARIABLE 3: People's experience a general sense of emptiness (survey response)

# Labeling

Project_Data$Lon_1 <- factor(Project_Data$Lon_1,
                             levels = c(1, 2, 3),
                             labels = c("yes", "more or less", "no"))

# Frequency table

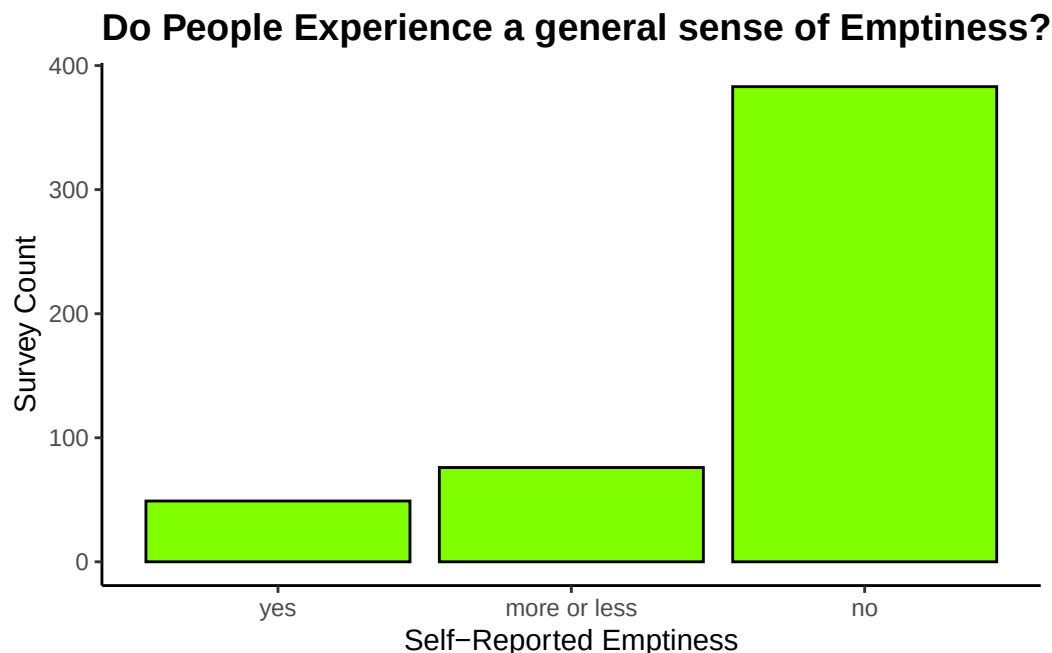
freq(as.ordered(Project_Data$Lon_1), plot = FALSE)
```

```
as.ordered(Project_Data$Lon_1)
```

	Frequency	Percent	Valid Percent	Cum Percent
yes	49	9.515	9.646	9.646
more or less	76	14.757	14.961	24.606
no	383	74.369	75.394	100.000
NA's	7	1.359		
Total	515	100.000	100.000	

```
# Bar graph

ggplot(data = subset(Project_Data, !is.na(Lon_1)), aes(x = Lon_1)) +
  geom_bar(fill = "chartreuse", color = "black") +
  xlab("Self-Reported Emptiness") +
  ylab("Survey Count") +
  ggtitle("Do People Experience a general sense of Emptiness?") +
  theme_classic() +
  theme(plot.title = element_text(size = 14, face = "bold"))
```



- **Univariate Analysis:** The first graph illustrates people's experiences of a general sense of emptiness. Overall, most respondents said no, while the fewest number of respondents said yes.
- **Lit Review motive:** I chose this variable (**people's sense of emptiness**) because it may contribute to experiences of social and emotional loneliness. If time spent nature is associated with lower loneliness on both social and emotional, we might also see a decrease in emptiness to those who spend more time in nature.

```
# VARIABLE 4: I miss having people around (survey response)

# Labeling

Project_Data$Lon_4 <- factor(Project_Data$Lon_4,
                             levels = c(1, 2, 3),
                             labels = c("yes", "more or less", "no"))

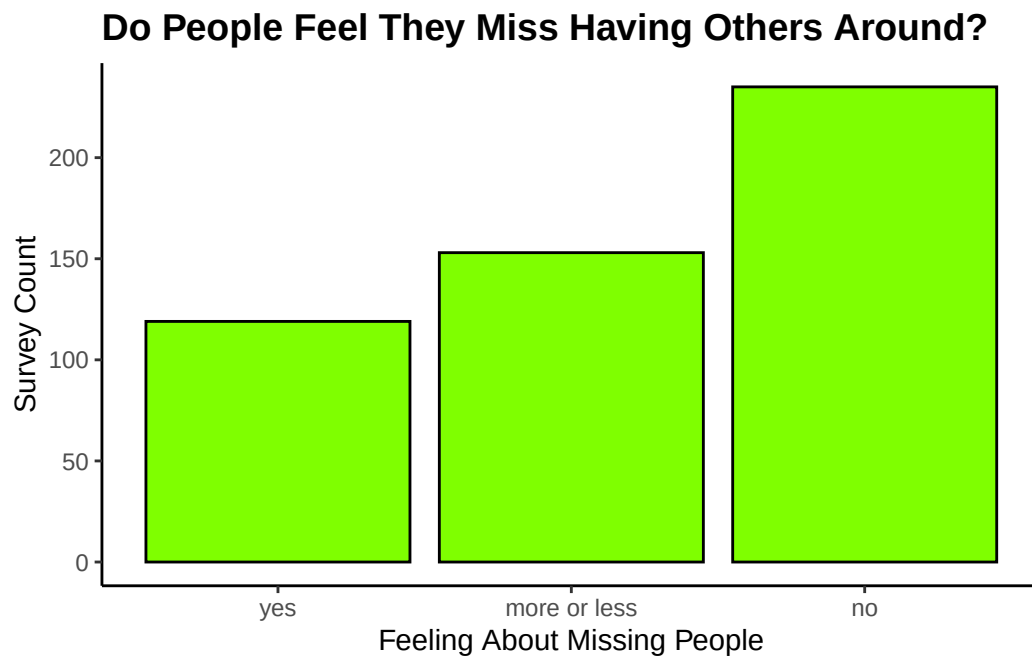
# Frequency table

freq(as.ordered(Project_Data$Lon_4), plot = FALSE)
```

```
as.ordered(Project_Data$Lon_4)
      Frequency Percent Valid Percent Cum Percent
yes           119   23.107         23.47      23.47
more or less   153   29.709         30.18      53.65
no            235   45.631         46.35     100.00
NA's             8    1.553
Total         515  100.000         100.00
```

```
# Bar graph

ggplot(data = subset(Project_Data, !is.na(Lon_4)), aes(x = Lon_4)) +
  geom_bar(fill = "chartreuse", color = "black") +
  xlab("Feeling About Missing People") +
  ylab("Survey Count") +
  ggtitle("Do People Feel They Miss Having Others Around?") +
  theme_classic() +
  theme(plot.title = element_text(size = 14, face = "bold"))
```



- **Univariate Analysis:** The second graph illustrates people's survey responses to whether individuals feel that they miss others in their lives. This is similar to the first graph. Most respondents said no, while the fewest number of respondents said yes.

- **Lit Review motive:** This variable (**missing social interaction**) could be important because social loneliness is being examined here. Comparing this to time spent in nature can help show whether nature can also regulate/help social loneliness as well.

```
# VARIABLE 5: I have high self-esteem (survey response)

# Labeling

Project_Data$SE_1 <- factor(Project_Data$SE_1,
                             levels = c(1, 2, 3, 4, 5),
                             labels = c("not very true of me", "2", "3", "4", "very true of me"),
                             ordered = TRUE)

# Frequency table

freq(as.ordered(Project_Data$SE_1), plot = FALSE)
```

```
as.ordered(Project_Data$SE_1)
      Frequency Percent Cum Percent
not very true of me      35   6.796      6.796
2                        69  13.398     20.194
3                       146  28.350     48.544
4                       193  37.476     86.019
very true of me          72  13.981    100.000
Total                   515 100.000
```

Lastly, I also chose this variable (**people's self-esteem**) because those who experience loneliness and spend little time in nature differ from those who don't feel loneliness and do spend time in nature. Perhaps those who do spend more time have higher levels of agreement to self-esteem compared to those who do not.

I may not add this to the poster at all.

3 Bivariate Analysis

```
# Bivariate 1: Correlation between age and time spent in nature

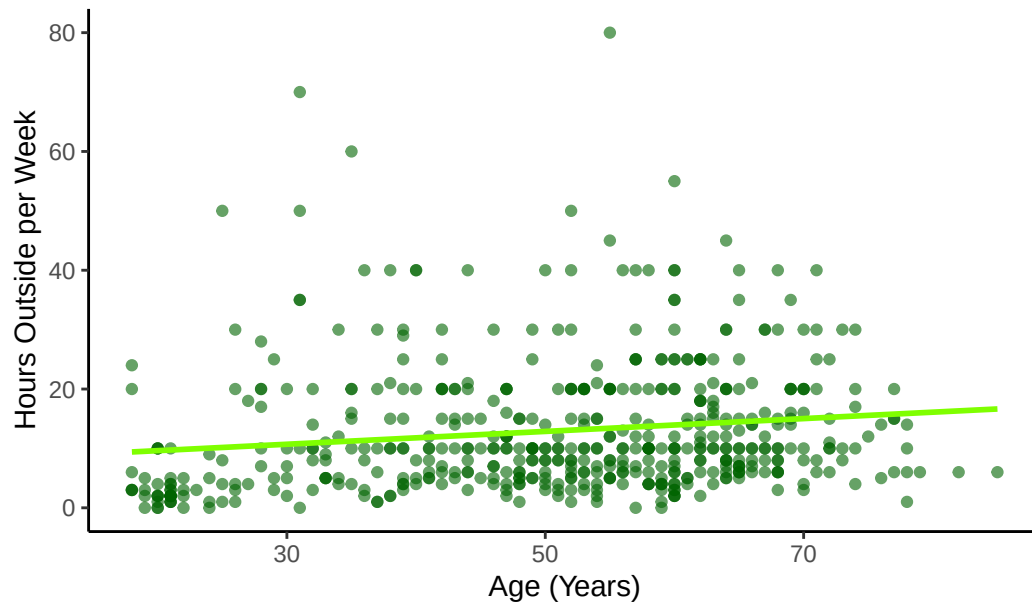
Project_Data <- subset(Project_Data, !is.na(D_Age) & !is.na(D_hours))

# Scatter plot

ggplot(Project_Data, aes(x = D_Age, y = D_hours)) +
  geom_point(color = "darkgreen", alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, color = "chartreuse") +
  xlab("Age (Years)") +
  ylab("Hours Outside per Week") +
  ggtitle("Does Age Influence Time Spent in Nature?") +
  theme_classic() +
  theme(plot.title = element_text(size = 14, face = "bold"))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

Does Age Influence Time Spent in Nature?



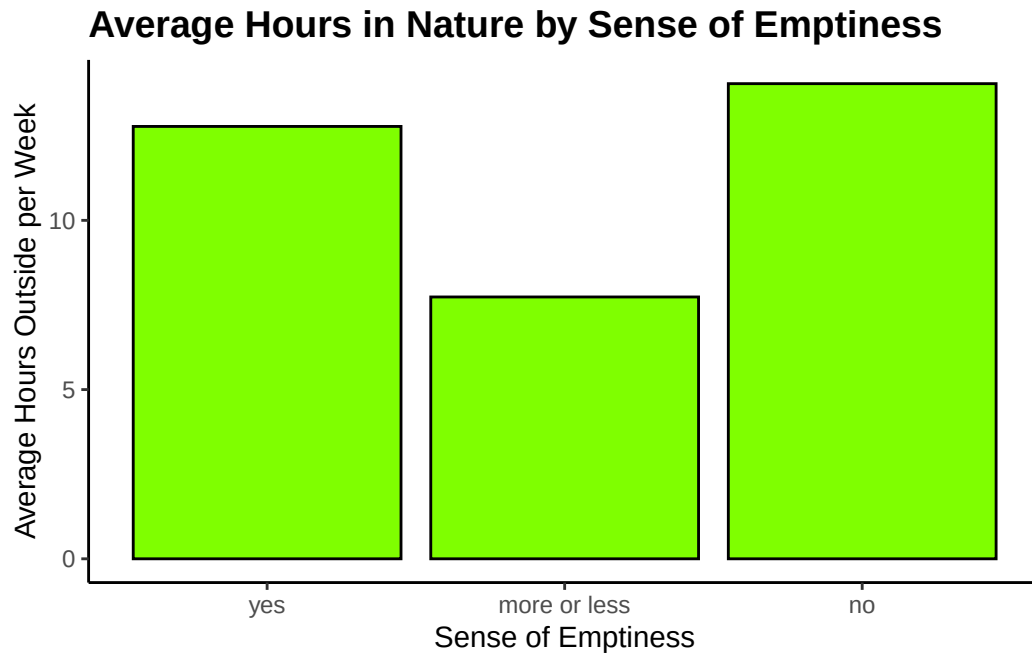
- The scatter plot shown here shows that there is a *slight* positive correlation between age and time spent outside hourly per week. The positive correlation shows that older individuals might spend a little bit more time in nature. However, when it comes to my research interest concerning the relationship between loneliness and nature, age does not seem to be a strong factor.

```
# Bivariate 2: Correlation between time spent in nature and general emptiness
```

```
summary_data <- Project_Data %>%  
  filter(!is.na(Lon_1) & !is.na(D_hours)) %>%  
  group_by(Lon_1) %>%  
  summarise(Avg_Hours = mean(D_hours))
```

```
# Bar graph  
ggplot(summary_data, aes(x = Lon_1, y = Avg_Hours, fill = Lon_1)) +  
  geom_bar(stat = "identity", fill = "chartreuse", color = "black") +  
  xlab("Sense of Emptiness") +  
  ylab("Average Hours Outside per Week") +  
  ggtitle("Average Hours in Nature by Sense of Emptiness") +  
  theme_classic() +  
  theme(plot.title = element_text(size = 14, face = "bold")) +  
  guides(fill = FALSE)
```

Warning: The `<scale>` argument of `guides()` cannot be `FALSE`. Use "none" instead as of ggplot2 3.3.4.



- The bar graph shown here illustrates the relationship between people's sense of emptiness, and their average hours spent in nature. Individuals who said **YES** to feeling empty showed a relatively high number of hours outside, and are like those who say **NO** to feeling empty as well. However, people who said **MORE OR LESS**, spent the least amount of time outside. They would mean that the relationship between a sense of emptiness and time spent in nature is not straightforward and there could be other factors influencing nature and loneliness.

4 Chi-Square Test of Independence

```
# Recode D_Gender into a factor with labels "man" and "woman"
Project_Data$D_Gender <- factor(Project_Data$D_Gender, levels = c(1, 2), labels = c("man", "woman"))

# Create a frequency table for D_Gender
gender_counts <- table(Project_Data$D_Gender)

# Print the counts
print(gender_counts)
```

```
man woman
143    370
```

```
# Men Chi-Square
Men_Data <- subset(Project_Data, D_Gender == "man")

# Remove missing data
Men_Data_Clean <- subset(Men_Data, !is.na(Nature_Hours_Group) & !is.na(Lon_1))

# Create the contingency table
table_men <- table(Men_Data_Clean$Nature_Hours_Group, Men_Data_Clean$Lon_1)

cat("Contingency table for men:\n")
```

Contingency table for men:

```
print(table_men)
```

	yes	more or less	no
Low (0-5)	4	11	18
Moderate (6-15)	1	8	46
High (16-30)	1	2	34
Very High (30+)	2	0	14

```
# Chi-Square Test
if (sum(table_men) == 0) {cat("No valid data in table for men.\n")} else {
  chisq_test_men <- chisq.test(table_men)
  cat("\n Chi-Square Test for Men:\n")
  print(chisq_test_men)}
```

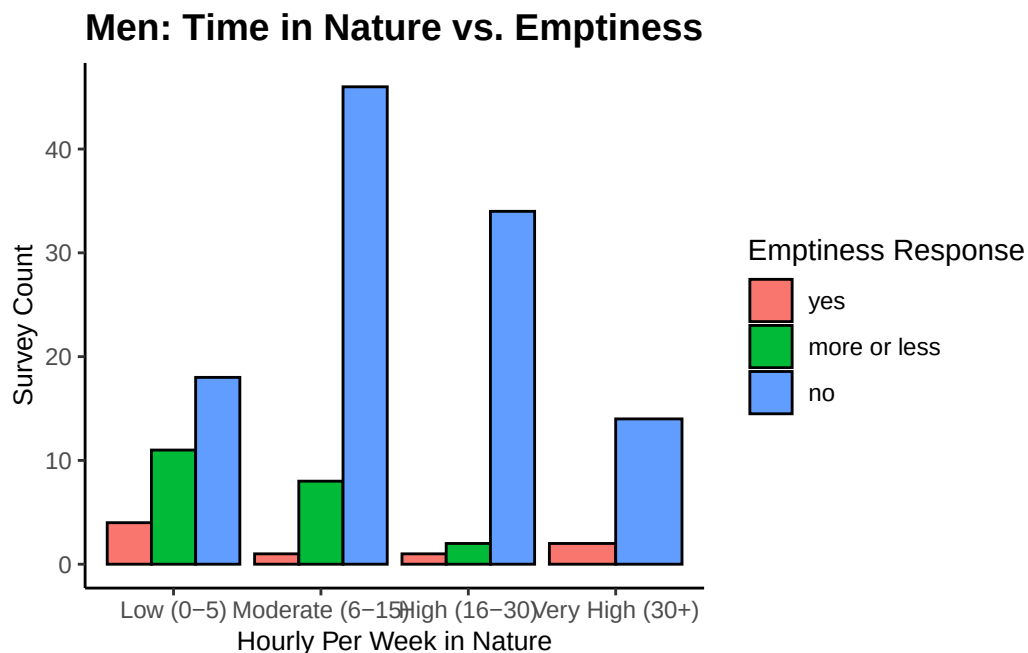
Warning in chisq.test(table_men): Chi-squared approximation may be incorrect

Chi-Square Test for Men:

Pearson's Chi-squared test

data: table_men
X-squared = 21.456, df = 6, p-value = 0.001519

```
# Bar Chart
ggplot(Men_Data_Clean, aes(x = Nature_Hours_Group, fill = Lon_1)) +
  geom_bar(position = "dodge", color = "black") +
  labs(title = "Men: Time in Nature vs. Emptiness",
       x = "Hourly Per Week in Nature",
       y = "Survey Count",
       fill = "Emptiness Response") +
  theme_classic() +
  theme(plot.title = element_text(size = 14, face = "bold"),
        axis.title.x = element_text(size = 10),
        axis.title.y = element_text(size = 10),
        axis.text.x = element_text(size = 9),
        axis.text.y = element_text(size = 9))
```



Chi-Square Test of Independence Analysis #1: Men

This analysis tested whether there is a relationship between mens time spent in nature hourly per-week (grouped into four categorical levels), and their reported sense of emptiness (“yes,” “more or less,” and “no”).

Analyzing the results, most men across all four categories reported not feeling empty. This shows a potential trend: the more time men spend time in nature each week, the less likely they are to report feelings of emptiness. However, in the “**Very High (30+)**” hours group, there was a modest increase in feelings of emptiness, showing that this relationship is not entirely linear and may be influenced by other factors.

The chi-square test resulted in a p-value of 0.001, making it statistically significant. This value is below the threshold of 0.05, **therefore we reject the null hypothesis**. Our null hypothesis was that there is no relationship between time spent in nature and symptoms of emptiness among men.

```
# Women Chi-Square
Women_Data <- subset(Project_Data, D_Gender == "woman")

# Remove missing data
Women_Data_Clean <- subset(Women_Data, !is.na(Nature_Hours_Group) & !is.na(Lon_1))

# Contingency Table
table_women <- table(Women_Data_Clean$Nature_Hours_Group, Women_Data_Clean$Lon_1)

cat("Contingency table for women:\n")
```

Contingency table for women:

```
print(table_women)
```

	yes	more or less	no
Low (0-5)	13	28	60
Moderate (6-15)	16	21	133
High (16-30)	8	6	66
Very High (30+)	2	0	12

```
# Chi-Square Test
if (sum(table_women) == 0) {cat("No valid data in table for women")}
else {
  chisq_test_women <- chisq.test(table_women)

cat("\n Chi-Square Test for Women:\n")
print(chisq_test_women)}
```

Warning in chisq.test(table_women): Chi-squared approximation may be incorrect

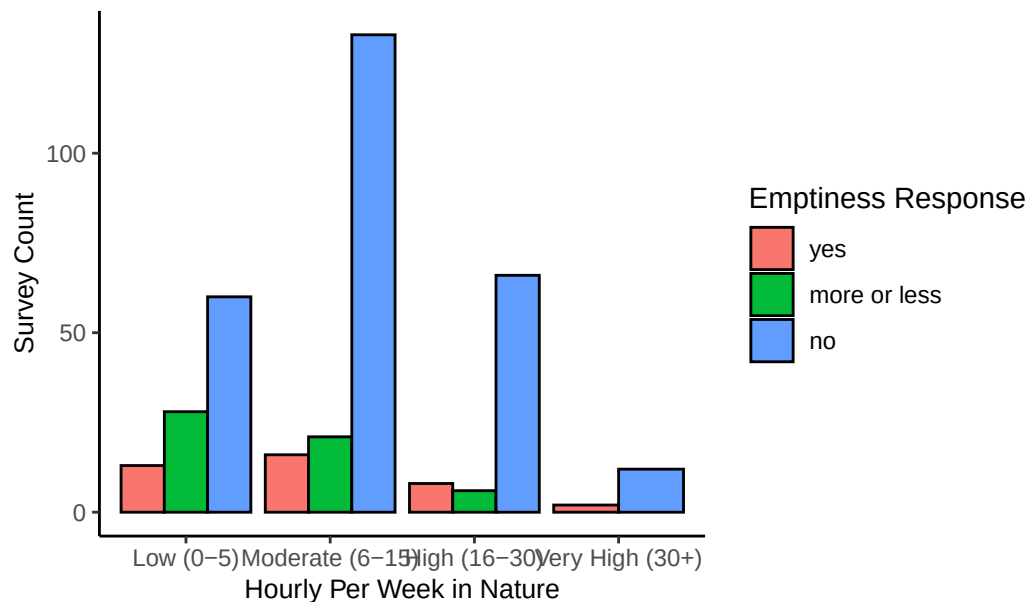
Chi-Square Test for Women:

Pearson's Chi-squared test

data: table_women
X-squared = 21.973, df = 6, p-value = 0.001225

```
# Bar chart
ggplot(Women_Data_Clean, aes(x = Nature_Hours_Group, fill = Lon_1)) +
  geom_bar(position = "dodge", color = "black") +
  labs(title = "Women: Time in Nature vs. Emptiness",
       x = "Hourly Per Week in Nature",
       y = "Survey Count",
       fill = "Emptiness Response") +
  theme_classic() +
  theme(plot.title = element_text(size = 14, face = "bold"),
        axis.title.x = element_text(size = 10),
        axis.title.y = element_text(size = 10),
        axis.text.x = element_text(size = 9),
        axis.text.y = element_text(size = 9))
```

Women: Time in Nature vs. Emptiness



Chi-Square Test of Independence Analysis #2: Women

A similar analysis was done for women to determine whether their time spent in nature is related to reported feelings of emptiness.

The test analysis yielded similar finding when compared to the previous test concerning men. Therefore, what was stated previously can also be stated here with women too.

In essence, besides rejecting the null hypothesis in favor of the alternative one, these two test support the idea that spending time in nature may reduce feelings of emptiness. This is further evidence as the p-value was the same for woman (0.001) as it was for men.

5 Single T-Test

```
# T-test
nature_hours_clean <- na.omit(Project_Data$D_hours)

t_test_result <- t.test(nature_hours_clean, mu = 5, alternative = "less")

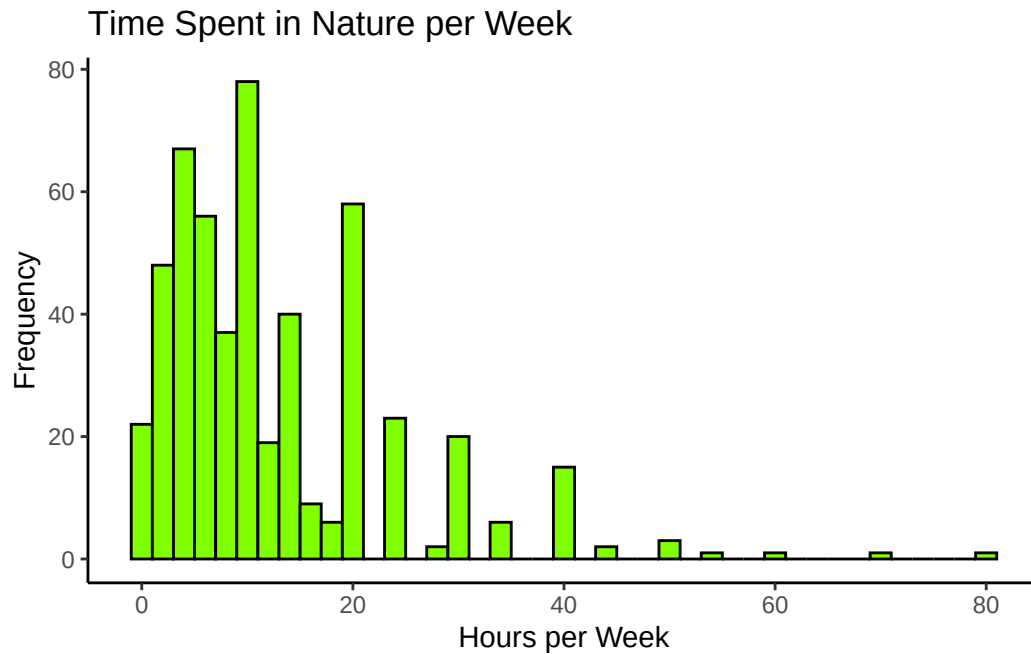
print(t_test_result)
```

One Sample t-test

```
data: nature_hours_clean
t = 16.338, df = 514, p-value = 1
alternative hypothesis: true mean is less than 5
95 percent confidence interval:
 -Inf 13.71495
sample estimates:
mean of x
12.9165
```

```
# Histogram
ggplot(Project_Data, aes(x = D_hours)) +
  geom_histogram(binwidth = 2, fill = "chartreuse", color = "black") +
  labs(title = "Time Spent in Nature per Week",
       x = "Hours per Week",
```

```
y = "Frequency") +  
theme_classic()
```



Null Hypothesis: People spend 5 hours or more per week in nature

Alt Hypothesis: People spend less than 5 hours per week in nature

This is a single T-test to see whether people spend less than 5 hours per week in nature.

This histogram shows that people spend more than 5 hours in nature, with the mean being around 13 hours per week. The data is skewed to the right, meaning more people report higher time in nature.

The T-test gave a p-value of 1. It is not statistically significant to support the alternative hypothesis. It would in fact support the null hypothesis.

This also suggests that people, on average, are getting a decent amount exposure to nature. It is a good amount for mental well-being.